

LexAI · Biweekly Progress Report

Week 3 – Week 4 (April 13 – April 26, 2026) · System Design Phase

Project ID	LEXAI-2026-SPRING	Phase	System Design (W3–W4)
Team Leader	Xiangyu Xiao	Submitted	April 26, 2026
Members	Baoyi Hu · Sirui Da · Qi Lin	Next Milestone	M3: Sprint 1 Complete — May 10

1. Overall Progress

Milestone M2 (Design Approved) was delivered **on schedule, April 26**. All five System Design sub-tasks (WBS 3.1–3.5) are closed. Cumulative hours in W3–W4: **118 h actual vs. 116 h planned (+1.7%)**. The team enters Development Sprint 1 on April 27 as planned.

2. Work Hours by Phase & Member

Work Package (WBS)	Plan h	Act h	Δ h	Xiangyu	Baoyi	Sirui	Qi Lin
3.1 Architecture Design	14	15	+1	6	4	3	2
3.2 RAG & Retrieval Design	14	14	0	2	8	2	2
3.3 Prompt / Output Design	12	13	+1	2	6	3	2
3.4 UI/UX Prototyping	10	10	0	1	2	5	2
3.5 Test Plan & Env. Setup	8	8	0	2	2	2	2
1.2/1.4 PM & Comms	8	8	0	5	1	1	1
4.1/4.6/4.7 Sprint 1 pre-work	0	10	+10	2	3	3	2
TOTAL	116	118	+2	20	26	19	13

3. Key Quantitative Achievements

Metric	Target / Baseline	Result (W3–W4)
Architecture components defined	—	23 components across 5 layers
RAG retrieval methods evaluated	Select 1	3 compared; hybrid BM25+dense selected (recall@5: 81%)
Prompt templates drafted (WBS 3.3)	≥4 scenarios	6 templates across 4 functional modules
UI prototype screens (Figma)	Core flows	11 screens, 3 user-flow paths, 14 reusable components mapped
Legal test scenarios (WBS 3.5)	≥15 (R2.2)	18 annotated scenarios finalised
Backend API endpoints (pre-work)	0 planned	9 routes scaffolded; CI pipeline passing
M2 review outcome	Approved	Approved Apr 26 · 0 blocking issues · 2 minor comments resolved

4. Schedule · Budget · Risk

Area	Status	Detail
Schedule	On Track	M2 delivered Apr 26 as planned; Sprint 1 starts Apr 27
Budget	On Track	¥700 spent of ¥3,500 (20%); Req. Analysis ¥300 + Design ¥400
Model access	Resolved	Tencent AI access confirmed; quota monitoring in place
Legal reviewer	Pending	Still TBD; PM to escalate by Apr 30 to avoid template-review delay

Xiangyu Xiao

Team Leader / PM · Architecture & Full-Stack Dev · 2351110@tongji.edu.cn · Apr 13 – Apr 26, 2026

Planned Hours	Actual Hours	Variance
20 h	20 h	0 h (0%)

1. Task Log & Deliverables

Task (WBS)	Hours	Key Deliverable / Quantitative Output
3.1 Architecture Design (lead)	6 h	System Architecture Doc v1.0 delivered; 5-layer diagram with 23 components (presentation, API gateway, service, retrieval, storage)
3.2 RAG Design (review)	2 h	Reviewed RAG strategy doc; signed off hybrid BM25+dense selection; approved latency budget (≤ 3 s retrieval + ≤ 2 s generation)
3.3 Prompt Design (review)	2 h	Reviewed all 6 prompt templates; raised 4 revision comments; all resolved before M2 review
3.5 Test Plan & Env.	2 h	Co-defined performance baselines: $P95 \leq 5$ s, system availability $\geq 95\%$ during demo
1.2 Progress Tracking	3 h	2 × weekly status reports issued; risk register updated (2 risks re-assessed; API quota risk kept at Medium)
1.4 Stakeholder Comms	2 h	M2 milestone review with Prof. Du on Apr 25; meeting minutes drafted and distributed within 24 h
4.1 Backend Scaffold (pre-work)	3 h	9 API endpoints stubbed (auth, routing, consultation, case, contract, docgen); auth middleware skeleton complete
TOTAL	20 h	

2. Highlights & Analysis

- Architecture doc approved at M2 with 0 blocking issues; 1 minor comment on data-flow notation resolved same day.
- Backend pre-work started 2 days ahead of Sprint 1 schedule; 9 route stubs reduce Sprint 1 ramp-up risk.
- Risk register: legal reviewer remains TBD — escalation email drafted, to be sent by Apr 30.
- Total variance: 0 h. All PM deliverables delivered on time with no scope changes.

3. Plan for Next Period (Apr 27 – May 10, Sprint 1)

- Lead WBS 4.1 Backend API & Services to completion: auth + routing + core service layer live by May 10 (planned 28 h).
- Coordinate Sprint 1 integration: ensure Baoyi's RAG pipeline and Sirui's frontend connect to API stubs by May 7.
- Resolve legal reviewer assignment by Apr 30; update risk register accordingly.
- Issue Sprint 1 status report at mid-point (May 3) and at M3 (May 10).

Submitted by: Xiangyu Xiao · April 26, 2026

Baoyi Hu

Model-Service Refinement & Output-Logic Tuning · 2353409@tongji.edu.cn · Apr 13 – Apr 26, 2026

Planned Hours	Actual Hours	Variance
22 h	24 h	+2 h (+9.1%)

1. Task Log & Deliverables

Task (WBS)	Hours	Key Deliverable / Quantitative Output
3.2 RAG & Retrieval Design (lead)	8 h	RAG Strategy Doc v1.0; 3 retrieval methods benchmarked on 50 queries: BM25 recall@5=68%, dense=74%, hybrid=81% → hybrid adopted
3.3 Prompt / Output Design (lead)	6 h	6 prompt templates drafted; avg. output consistency score 82% across 20 internal test cases; citation-format spec defined
3.1 Architecture — RAG layer	4 h	Specified LLM ↔ retriever interface contract; latency budget: ≤3 s retrieval + ≤2 s LLM generation = ≤5 s total (P95)
3.5 Test Plan — benchmark criteria	2 h	Hallucination benchmark: 60 seed Q&A pairs compiled from PRC Civil Code for audit (target ≤8% unsupported-claim rate)
4.6 KB / RAG pre-work	3 h	Initial corpus: 120 PRC Civil Code articles chunked (avg. 280 tokens/chunk, target 256–512); draft vector index created
1.4 Team comms	1 h	Attended 2× weekly syncs; presented RAG strategy at M2 review
TOTAL	24 h	

2. Highlights & Analysis

- Hybrid BM25+dense retrieval selected based on empirical recall@5 comparison on 50 internal queries — 13 pp improvement over BM25 alone.
- 120 Civil Code articles chunked and indexed as proof-of-concept; full 1,260-article corpus indexing scheduled for Sprint 1 (WBS 4.6, 30 h).
- 6 prompt templates cover all 4 functional modules; avg. consistency 82% — target ≥85% after iteration in Sprint 1.
- Overrun of +2 h (+9.1%) due to additional prompt-iteration cycles on contract-review templates; no schedule impact on M2 delivery.
- 60 benchmark Q&A pairs ready for hallucination audit in WBS 5.2; on track for ≤8% unsupported-claim target.

3. Plan for Next Period (Apr 27 – May 10, Sprint 1)

- Complete Legal KB indexing: ingest all 1,260 PRC Civil Code articles; achieve recall@5 ≥ 75% on benchmark set (WBS 4.6, ~15 h).
- Build and integrate RAG query pipeline with backend API stubs; end-to-end latency test targeting P95 ≤ 5 s.
- Begin Consultation Module (WBS 4.2): citation output pipeline, boundary-reminder injection (~10 h).
- Iterate prompt templates based on Sprint 1 integration results; target consistency score ≥ 85%.

Submitted by: Baoyi Hu · April 26, 2026

Sirui Da

Workflow Organisation & Interaction Polishing · 2352288@tongji.edu.cn · Apr 13 – Apr 26, 2026

Planned Hours	Actual Hours	Variance
19 h	19 h	0 h (0%)

1. Task Log & Deliverables

Task (WBS)	Hours	Key Deliverable / Quantitative Output
3.4 UI/UX Prototyping (lead)	9 h	Figma prototype: 11 screens across 4 modules (consultation, case analysis, contract review, doc generation); 3 complete user-flow paths
3.3 Prompt Design — interaction	3 h	Boundary-reminder logic added to 4 consultation templates; 3 edge-case interaction scripts drafted and reviewed
3.1 Architecture — frontend layer	3 h	Frontend component map: 14 reusable React components identified; estimated 20% reduction in Sprint 2 frontend dev time
3.5 Test Plan — UX scenarios	2 h	18 legal scenarios validated for UI-flow coverage; all 4 functional modules fully covered by at least 3 scenarios each
4.7 Frontend pre-work	1 h	React project scaffolded; design tokens (colours, typography, spacing) committed to shared repo; ready for Sprint 1
1.4 Team comms	1 h	Attended 2× weekly syncs; presented Figma prototype at M2 review; 2 nav changes requested and resolved same day
TOTAL	19 h	

2. Highlights & Analysis

- 11 Figma screens delivered on schedule; M2 feedback: 2 minor navigation adjustments, both resolved within the same day (Apr 25).
- 14 reusable React components mapped — estimated 20% time saving in WBS 4.7 (38 h budgeted), equivalent to ~7-8 h saved in Sprint 2.
- 18 legal test scenarios finalised, exceeding the minimum of 15 required by WBS 2.2 / R2.2.
- Zero variance against plan (19 h planned, 19 h actual); all 4 UX deliverables closed before M2 review.
- Boundary-reminder logic for consultation module agreed with Baoyi; ensures R1.1 compliance (cited, bounded answers).

3. Plan for Next Period (Apr 27 – May 10, Sprint 1)

- Start WBS 4.7 Frontend Web App: implement core layout, navigation shell, and consultation chat interface (~16 h).
- Connect consultation UI to Xiangyu's API stubs; achieve end-to-end UI ↔ API integration for consultation flow by May 8.
- Begin WBS 4.3 case-analysis screen layout in parallel with Baoyi's module development (~6 h).
- Conduct first internal usability walkthrough with team by May 9; document ≥3 UX issues for backlog.

Submitted by: Sirui Da · April 26, 2026

Qi Lin

Interface Integration · Testing · Deployment · Delivery · 2352609@tongji.edu.cn · Apr 13 – Apr 26, 2026

Planned Hours	Actual Hours	Variance
13 h	13 h	0 h (0%)

1. Task Log & Deliverables

Task (WBS)	Hours	Key Deliverable / Quantitative Output
3.5 Test Plan & Env. Setup (lead)	6 h	Test Plan v1.0: 5 test types, 47 test cases defined (18 functional R1.1–R1.4, 12 non-functional R2.1–R2.4, 9 security R2.2, 8 integration R1.5)
3.1 Architecture — deployment view	2 h	Deployment architecture diagram: 3-tier (React frontend / FastAPI gateway / PostgreSQL + Chroma vector store); infra spec documented
2.3 Evaluation criteria (final review)	1 h	MOV metrics doc reviewed and confirmed: P95 ≤ 5 s and availability ≥ 95% established as measurable and testable
4.1 Backend / CI pre-work	2 h	GitHub Actions CI pipeline configured: build + lint workflow; first automated build passed Apr 24 with 0 lint errors
4.6 KB environment setup	1 h	Chroma vector DB installed and verified: insert/query latency < 80 ms on 1,000-vector dataset (within ≤2 s retrieval budget)
1.4 Team comms	1 h	Attended 2× weekly syncs; shared deployment checklist with team; coordinated environment config with Baoyi
TOTAL	13 h	

2. Highlights & Analysis

- Test plan covers 47 cases across 5 dimensions; all linked to RBS requirements (R1.1–R1.5, R2.1–R2.4, R3.1).
- Chroma DB query latency verified at <80 ms on 1,000 vectors — well within the 2 s retrieval budget set by Baoyi; no DB bottleneck risk.
- CI pipeline live: first passing build Apr 24; lint step catches issues before review, reducing integration friction in Sprint 1.
- Zero variance against plan (13 h planned, 13 h actual); all deliverables submitted before M2.
- Deployment architecture covers all components in system architecture doc (WBS 3.1); no gaps identified.

3. Plan for Next Period (Apr 27 – May 10, Sprint 1)

- Start WBS 5.1 Unit & Integration Testing: write ≥30 unit tests covering backend API and RAG query pipeline; maintain CI green throughout Sprint 1.
- Finalise demo environment deployment (WBS 6.1): provision campus server, deploy backend + frontend stubs, smoke-test by May 9.
- Coordinate integration testing with Baoyi (RAG pipeline) and Sirui (frontend) by May 8; target 0 critical integration defects at M3.
- Update test plan to cover Sprint 1 modules (WBS 4.2 Consultation, WBS 4.6 RAG pipeline); add 10+ new test cases.

Submitted by: Qi Lin · April 26, 2026